Jaloliddin Ismailov

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Education

Eötvös Loránd University (ELTE) — **Budapest, HU**
BSc in Computer Science (GPA: 4.6/5.0; Member of Central Asian Society) — *Expected: Jun 2028*

Experience

BME Solar Boat Team — **Budapest, HU**
Firmware & Embedded Systems Engineer — *Feb 2026 – Present*

- Write firmware that controls the boat's throttle on an ESP32, in C and FreeRTOS, and tuning the low-level I2C/SPI drivers so the controls respond instantly and never freeze up.
- Built the system that collects and combines the boat's sensor readings, with bit-level error checking and raw ADC values scaled into insightful analysis. This lets the team watch the boat live and catch failures in the lab through Hardware-in-the-Loop (HIL) testing, instead of finding out on the water.

CERN — **Remote / Geneva, CH**
Software Engineering Research Fellow (LHCb) — *Nov 2025 – Jan 2026*

- Wrote algorithms reconstructing muon tracks through the LHCb detector in C++ CUDA on terabytes of sensor data, using ML and data structures.
- Automated the team's build and deploy process with Docker and CI/CD pipelines to ensure the same code runs the same way everywhere. Professors stopped losing hours to manual setup and could ship updates faster.

Projects

Assistive Smart Glasses — **BME Robotics Hackathon**
Hardware & Edge AI Developer [Video Demo] — *Jun 2026*

- Built with a team of four smart glasses that help the wearer navigate, running computer vision and TinyML models directly on a tiny ESP32-CAM. Did the processing on the device instead of sending video off to a server, decreasing latency. Matters a lot when someone's relying on it to move around safely.

Flood-Risk Monitoring System — **CASSINI Hackathon Budapest**
Backend Developer [Video Demo] — *Apr 2026*

- Built with a team of three the FastAPI backend for a water-use inspection dashboard for Hungarian farms that turns three separate satellite/GPS data sources into a single risk score inspectors can check per field.
- Combined Sentinel-2 satellite imagery – to estimate how much water a crop needs from its greenness – with Sentinel-1 radar to check actual soil moisture. Then used Galileo's OSNMA signal to verify that a water-meter reading really came from the claimed location at the claimed time.

Extracurriculars

European Young Engineers — **Remote**
Regional Coordinator — *Mar 2026 – Present*

- Represent thousands of young engineers across Europe at UN climate negotiations, advocating for sustainable development and the role of technology in climate action.

FLEX Program — **Minnesota, USA**
Exchange Student at Prior Lake High School — *Sep 2023 – Mar 2024*

- Won a competitive, fully-funded scholarship from the U.S. Department of State to study in the U.S. for an academic year.
- Completed 50+ hours of community service and led International Students' Club initiatives bringing students from different backgrounds together.